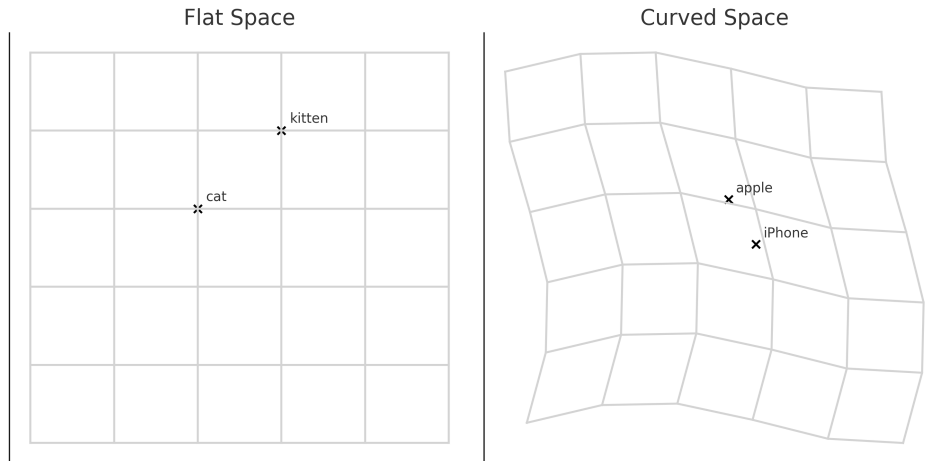


Seeing Inside the Black Box

A Geometric Perspective on AI

Saar Hamam
Independent Researcher
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Visualization 1: Flat vs. Curved Space — A warped embedding shifts meaning: "cat" and "kitten" stay near on flat space, while "apple" and "iPhone" pull closer in curved geometry.

1. From Words to Shapes: The Landscape of Meaning

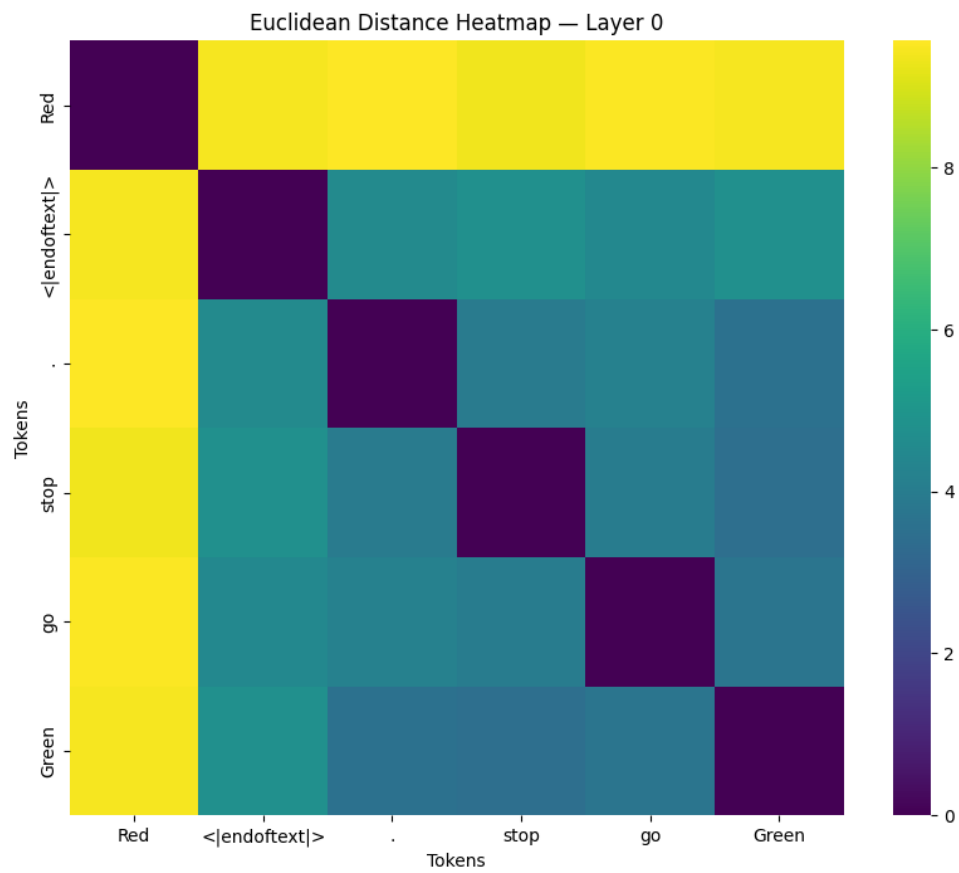
Imagine every word you know living in a vast invisible space. Words with similar meanings — like *cat* and *kitten* — live close together. Unrelated words — like *pencil* and *galaxy* — are far apart. This terrain is called embedding space. Each layer of an AI model reshapes this terrain, refining how meaning is understood.

2. Curvature: How the Model Bends Meaning

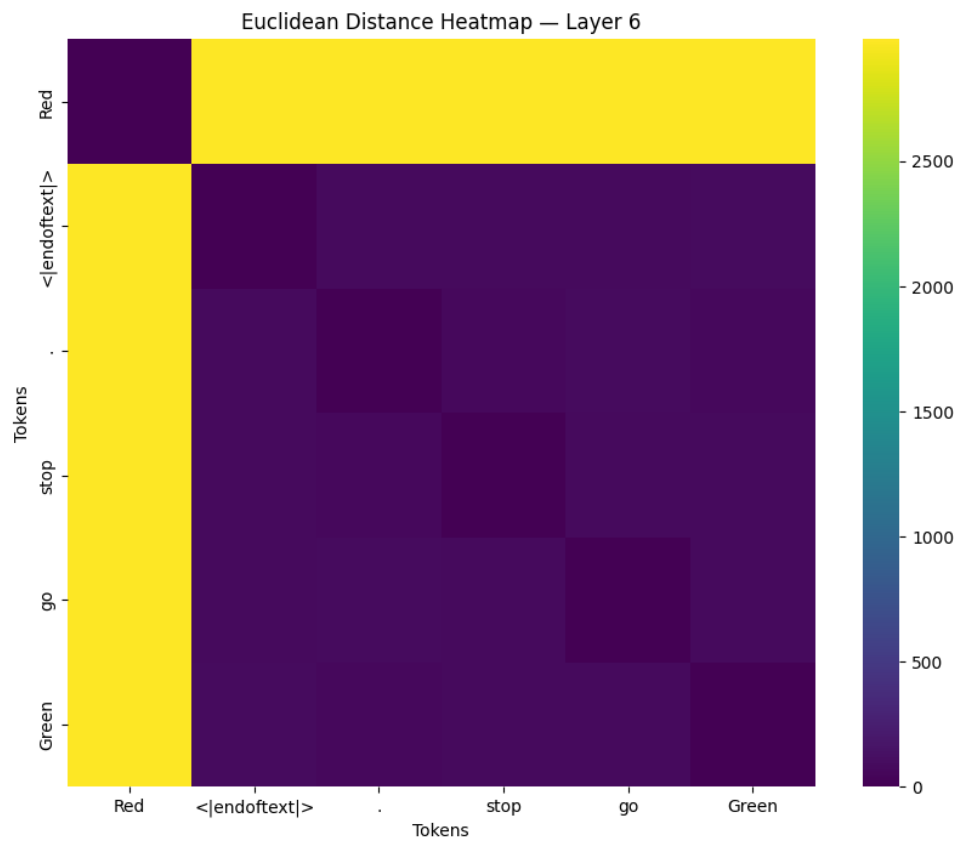
Picture that space as a rubber sheet. As context forms, the model bends the sheet: clustering ideas like *apple* and *iPhone*, while distancing unrelated ones. This curvature helps the model represent non-obvious connections.

Using a model only through its input-output surface flattens this geometry. It's like navigating mountains using a flat tourist map: technically usable, but disconnected from the terrain beneath.

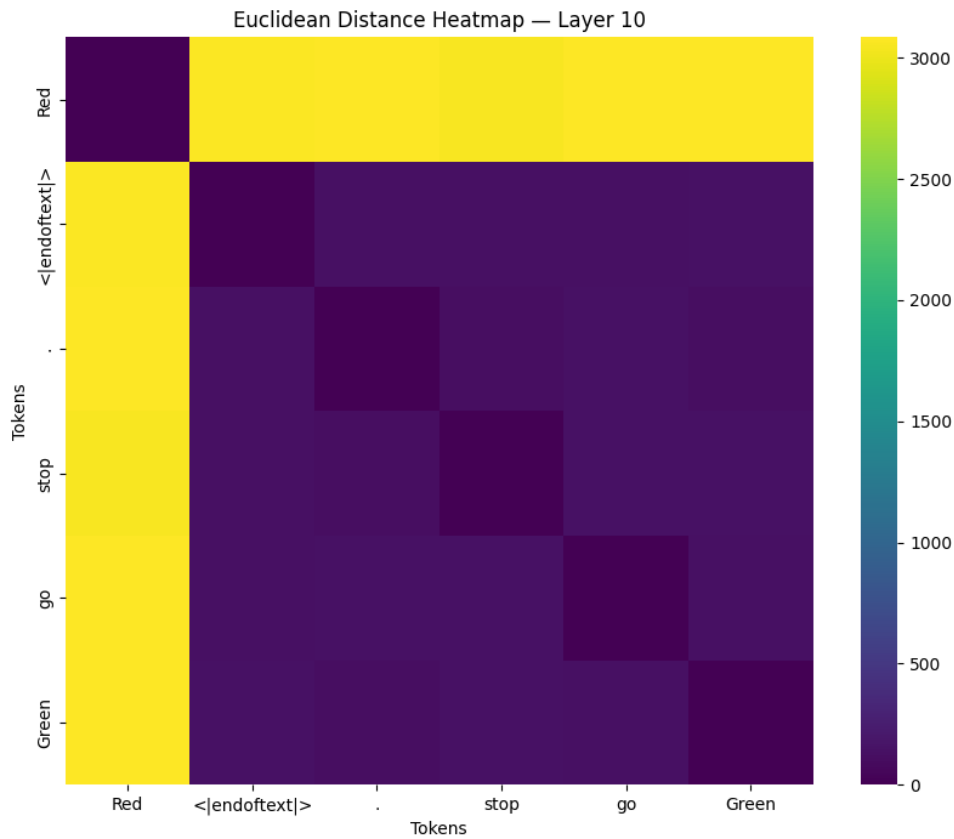
3. Why Euclidean (Flat) Distance Breaks



Euclidean Distance — Layer 0 — Distances reflect intuitive lexical proximity.



Euclidean Distance — Layer 6 — Distances begin collapsing. Signal is lost in magnitude scaling.



Euclidean Distance — Layer 10 — Semantic structure becomes unreadable. Outlier domination and saturation.

As information propagates through the network, simple Euclidean distance fails to capture meaning. While it works in early layers, inner layers distort representations such that:

- Distances explode or vanish (scale-sensitive)
- Clustering does not reflect semantic similarity
- Outliers dominate the geometry

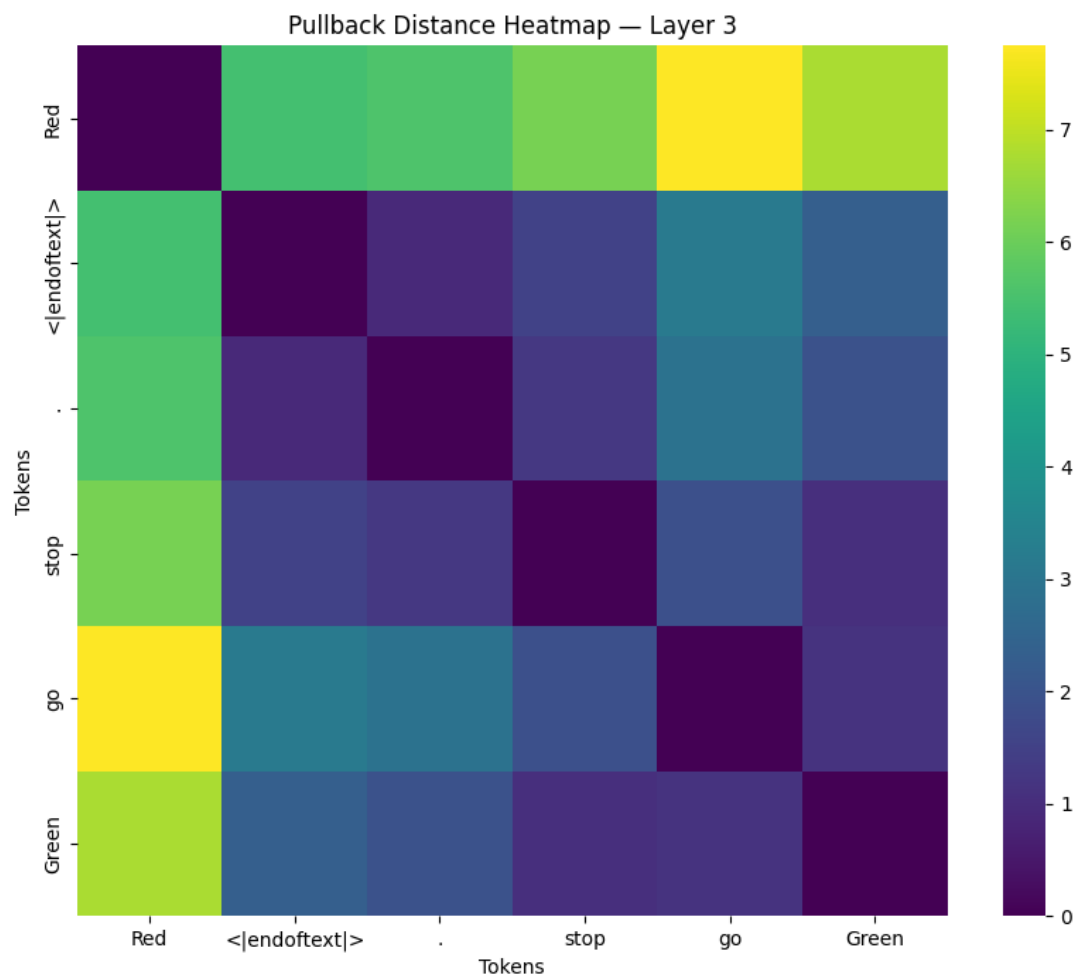
Pullback metrics, however, adapt to the model’s non-Euclidean structure. They measure local curvature and semantic deformation directly, revealing:

- True conceptual boundaries
- Layer-specific transitions
- Stable geometry across depth

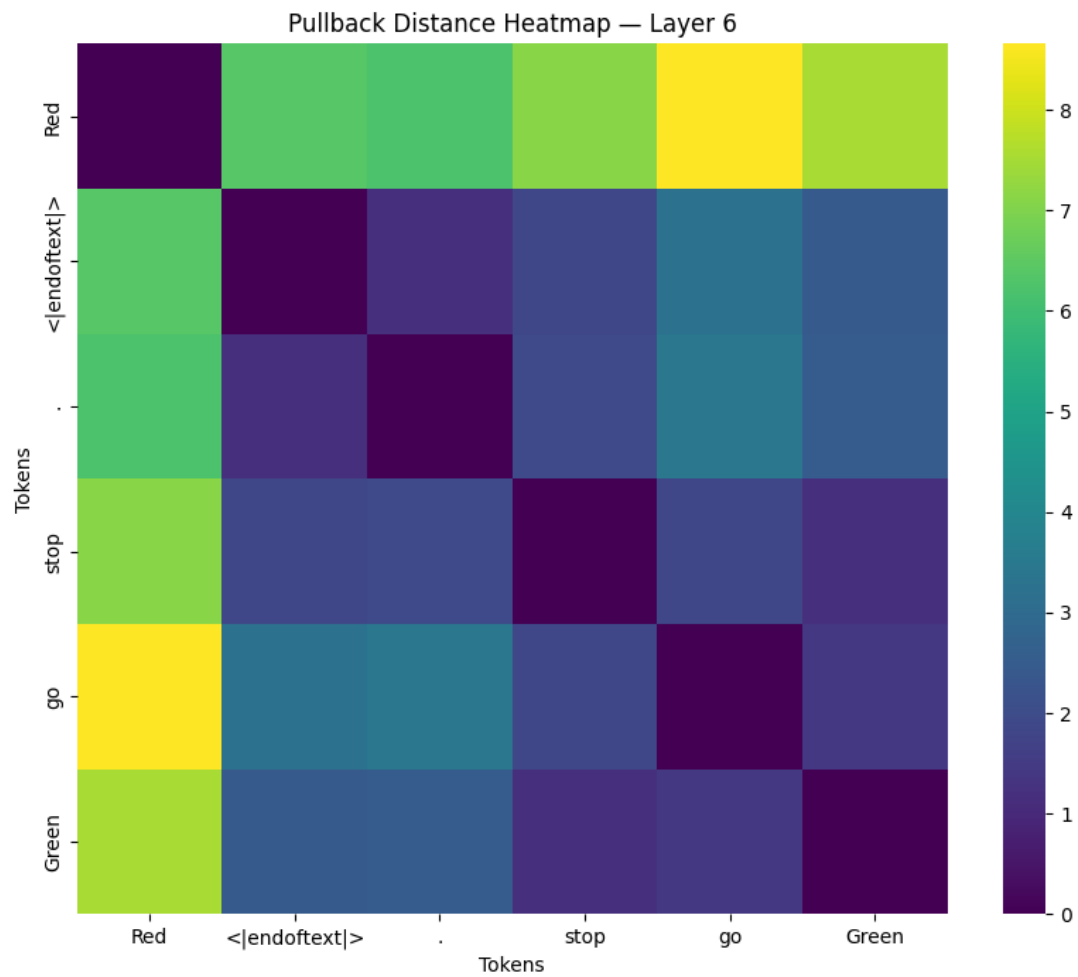
4. Cartographic Pullbacks: Mapping Across Layers



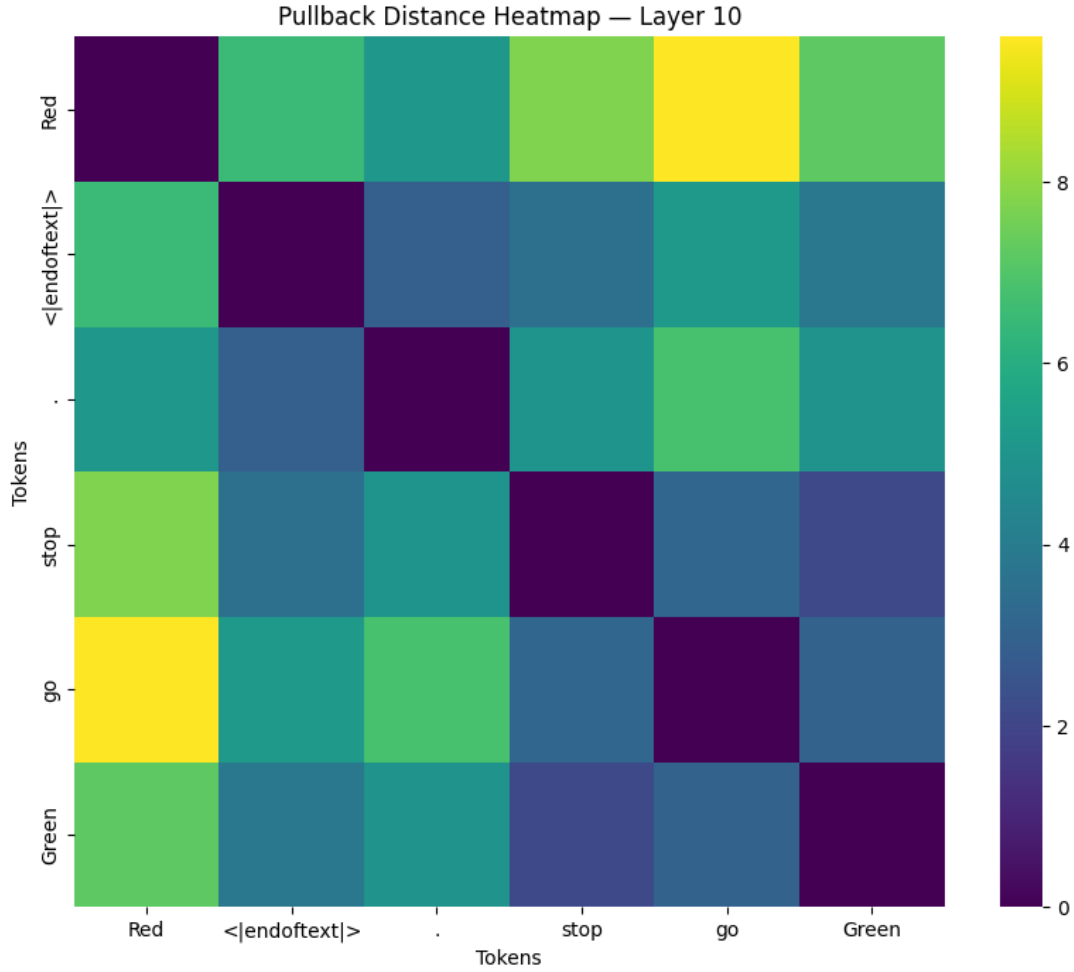
Layer 0 (Input) Pullback Distances — The geometry is shallow and token-based. Distances reflect surface-level dissimilarity rather than meaning. There is little semantic structure; “Red”, “Green”, “stop”, and “go” are not yet functionally grouped.



Layer 3 Pullback Distances — Early semantic shaping begins. The model starts grouping “Red” with “stop” and “Green” with “go”, reflecting emerging contextual sensitivity. The space is curved, but still diffuse.



Layer 6 Pullback Distances — Midway through the network, structure sharpens. Contextually relevant token pairs pull closer. The model’s internal geometry aligns with latent semantic fields. Distances encode learned associations.



Layer 10 Pullback Distances — The model has committed to a semantic interpretation. Distance collapses occur where concepts fuse, and sharp curvature emerges at decision boundaries. Pullback geometry now mirrors model certainty.

By pulling back geometric metrics from output space to token space, we create dynamic heatmaps of where meaning forms, shifts, or concentrates. These “maps” reveal:

- Which tokens ground the model’s understanding
- Where transitions between concepts emerge
- How curvature evolves across layers

5. Conclusion: From Mystery to Meaning

This geometric lens does not decode every aspect of a model’s internals. But it provides a rigorous and intuitive way to understand it: through curvature, distance, sensitivity, and semantic flow. We begin not only to use the black box, but to *see* inside it.